`Semigroups` — Zero to Hero — Worksheet

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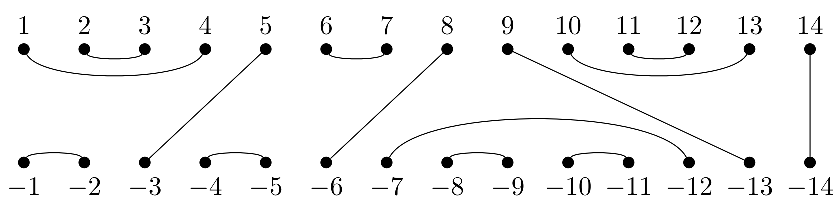
This worksheet contains some exercises to be completed in `GAP` with the use of the `Semigroups` package as part of the GAP workshop session of NBSAN 40. The problems have been written in a deliberately terse way to encourage attendees to experiment with the package and explore its associated manual: `https://semigroups.github.io/Semigroups/doc/chap0_mj.html`.

As well as inspecting the package manual, please also feel free to ask questions to James, Reinis or Joe!

1. Let J_n denote the Jones monoid of degree n, and

$x =$ 

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   (a) How many elements does J_{14} contain?
   (b) How many idempotent elements does J_{14} contain?
   (c) How many idempotent elements does the Green's $\mathcal{D}$-class of x in J_{14} contain?
   (d) How many idempotent elements with 7 in the domain does the Green's $\mathcal{D}$-class of x in J_{14} contain?